

Reading a magnet's Hamiltonian term by term: two-dimensional terahertz spectroscopy as vertex spectroscopy of TmFeO_3

Working draft¹

¹ —

(Dated: August 6, 2026)

One-dimensional spectroscopy measures the *spectrum* of a Hamiltonian; two-dimensional coherent spectroscopy (2DCS) measures its *vertices*—the individual nonlinear terms that convert one coherence into another. We show that in the rare-earth orthoferrite TmFeO_3 this correspondence can be made one-to-one and, for every observed feature, explicit. Time-reversal parity and space-group irrep bookkeeping reduce all symmetry-allowed Fe–Tm couplings to exactly two conversion vertices; the mirror at the Tm 4c site then assigns each light polarisation to one block of the crystal-field qutrit, fixing what each channel can excite and what it can radiate. Driven by the measured single-cycle waveform, with every damping rate set by a measured linewidth and every emission weight by a measured oscillator strength, one Hamiltonian reproduces both cross-polarised inventories without retuning and four of six same-polarised amplitudes within 15%. We give the pathway of every peak in all four channels—which vertex permits it, which coherence occupies the delay axis, which multipole radiates, and what would falsify the assignment—and show that the residual discrepancy reduces to a single dynamical ratio.

I. INTRODUCTION

Linear response cannot distinguish two Hamiltonians with the same normal modes: absorption measures pole positions and dipole weights, and any coupling that merely hybridises modes is absorbed into the mode basis. What it can never see is a *conversion*: a coherence prepared on one transition and re-emitted on another. That is precisely what a cross peak in 2DCS certifies [1, 2], and a conversion requires a specific anharmonic vertex. A cross peak at (ω_τ, ω_t) is therefore not a feature to be fitted but a *measurement of one coupling term*: the delay axis names the stored coherence, the detection axis the emitter, and the signed side constrains the pathway order. If symmetry makes the space of candidate vertices finite—and in a crystal it does—a polarisation-resolved 2D atlas can be read as the Hamiltonian itself.

TmFeO_3 is an unusually clean testbed because two qualitatively different subsystems share one THz octave. Below the spin reorientation the Fe sublattice orders in Γ_2 (F_x, C_y, G_z) [3–5], with quasi-ferromagnetic and quasi-antiferromagnetic magnons at $q_{\text{FM}} = 0.38$ and $q_{\text{AFM}} = 0.90$ THz, while the three lowest Tm^{3+} crystal-field (CEF) singlets form a qutrit with $E_{12} = 0.50$, $E_{23} = 0.70$, $E_{13} = 1.20$ THz. A single-cycle pulse covers all five modes, and measurements in the cross- and same-polarised channels of two field geometries return distinct, reproducible inventories.

II. MODEL, AND WHERE IT IS EXACT

The Fe^{3+} ($S = 5/2$) sublattice is classical, with the calibrated orthoferrite Hamiltonian (exchange, single-ion anisotropy, Dzyaloshinskii–Moriya) reproducing q_{FM} , q_{AFM} and Γ_2 . Each Tm^{3+} is truncated to its three lowest singlets and carried as the Gell-Mann vector $\lambda^a = \langle \Lambda^a \rangle$.

Because every single-ion term is linear in the Λ^a , the classical $\vec{\lambda}$ equation of motion *is* the exact von Neumann equation of the driven, damped qutrit, and a Boltzmann ensemble over level seeds is the exact thermal density matrix [SM, Sec. S1]. The Fe–Tm sector is built by projecting $\mathbf{S}^T \mathbf{A} \mathbf{J}$ into the qutrit in a transported gauge over the four Tm sublattices [SM, Secs. S2–S3]. All spectra use the measured pump (spectrum peaked at 0.53 THz, no DC), measured linewidths as damping rates, and measured oscillator strengths as emission weights. Peaks are identified blindly as 2D local maxima with prominence ≥ 1.5 ; the τ -averaged signal is subtracted at each detection time, removing the $\omega_\tau = 0$ row.

III. TWO VERTICES SURVIVE

A conversion vertex must (R1) reach the bright coordinate, (R2) be sourced by the pumped mode, (R3) be time-reversal even, (R4) be Γ_1 -allowed in the ordered state, (R5) carry the frequency mismatch, and (R6) engage both pulses to survive $M_{\text{NL}} = M_{AB} - M_A - M_B$. These exhaust the coupling space [SM, Sec. S4]. The static bilinear $K^- S \lambda$ fails threefold—linear in S it hybridises but cannot convert, its channels are Γ_1 -cancelled to machine precision, and its one allowed component re-orientates the ground state—and the quadratic families W_4 , W_1^{xy} , W_6 , κ^E are exactly inert. What remains is

$$H_W = W_1^{yy} S_y^2 \lambda^1, \quad H_\kappa = \kappa_{1y}^B B_y(t) S_y \lambda^1, \quad (1)$$

together with the static hybridisation $W_1^{xz} S_x S_z \lambda^1$ and the population channel $W_3 S_\alpha S_\beta \lambda^3$. Every peak below is built from these.

IV. SYMMETRY ASSIGNS EACH POLARISATION ONE BLOCK

The Tm 4c site carries the mirror m ($z \rightarrow -z$), under which $d_{x,y}$ are even and d_z odd, while m_z is even and $m_{x,y}$ odd. Two exact statements then fix the entire readout. First, time reversal: the CEF eigenstates are nondegenerate singlets and may be chosen real, in which basis J_α is imaginary-antisymmetric, so PJP has support *only* on $\lambda^{2,5,7}$ —the magnetic moment is rigorously off-diagonal, with no permanent component. The electric dipole, being $\mathcal{T}$ -even and real-symmetric, occupies exactly the complement $\lambda^{1,3,4,6,8}$, which *does* include the diagonal generators $\lambda^{3,8}$. Second, the mirror decides whether those diagonals survive: $\langle i|d_z|i \rangle = 0$ because d_z is odd, whereas $\langle i|d_x|i \rangle$ is allowed. The projected moment matrix—an output of the PJP construction, not an input—has λ^2 carrying only μ_z and $\lambda^{5,7}$ only $\mu_{x,y}$, which forces $|1\rangle, |2\rangle$ to share a parity and $|3\rangle$ to be opposite. Every coupling follows:

transition	magnetic	electric	owned by
$1 \leftrightarrow 2$	$m_z (\lambda^2)$	$d_x (\lambda^1)$	$(E \parallel a, H \parallel c)$
populations	—	$d_x (\lambda^3, \lambda^8)$	$(E \parallel a, H \parallel c)$
$1 \leftrightarrow 3$	$m_x (\lambda^5)$	$d_z (\lambda^4)$	$(E \parallel c, H \parallel a)$
$2 \leftrightarrow 3$	$m_x (\lambda^7)$	$d_z (\lambda^6)$	$(E \parallel c, H \parallel a)$

The population row is the one asymmetry between the two polarisations, and it is forced: both operators owned by $(E \parallel a, H \parallel c)$ are mirror-*even*, so that analyser setting reads the CEF level populations directly through d_x , while both operators owned by $(E \parallel c, H \parallel a)$ are mirror-*odd* and are therefore purely off-diagonal. A population carries no oscillation at a transition frequency, so this term radiates only near $\omega_t = 0$; it is a rectified, not a resonant, channel. Reciprocity—one operator per polarisation, driving and radiating alike—then implies that *cross-polarised detection reads exactly the block the pulse did not drive*. This is the structural reason cross-polarisation isolates conversion, and it carries two independent checks: it predicts the geometry-B subalgebra closure, where the c -directed drive touches only μ_z^2 so that λ^{4-7} vanish identically (verified: exactly 0, against $\lambda^{1,2,3} \sim 2.6 \times 10^{-2}$); and in geometry A $F_z \equiv 0$ to 10^{-15} , so the Fe sublattice is radiatively silent in that cross channel and anything seen there at a magnon frequency must be Tm.

Two effects extend this. Below T_N the Γ_2 order parameter is m_z -odd and breaks the site mirror, inducing a c -axis coupling d_z^{eff} to $1 \leftrightarrow 2$ that parity would forbid; this is what lets $E \parallel c$ create the E_{12} coherence dominating the geometry-A delay axis. And the 4c site has no inversion (Fe at 4b does), so the emitted moment may carry a quadratic term,

$$m_z = \mu \lambda^2 + \beta \lambda^1 \lambda^2, \quad (2)$$

$\mathcal{T}$ -odd as a magnetisation must be, radiating at $2\omega_{12}$. The resulting detection operators are collected in Table I.

Two consequences are worth stating before the pathways. The geometry-A cross and geometry-B same-polarised channels analyse the *same* output polarisation, so they are not independently adjustable: one operator must explain both. And because λ^1 and λ^2 are quadratures of the same $1 \leftrightarrow 2$ coherence, their magnitude spectra coincide to 10%; varying w_{E1} over a factor of 50 moves every geometry-A cross amplitude by at most 0.04 once β is refitted. The atlas is thus insensitive to the $E1/M1$ balance everywhere except one line—the rectified geometry-B feature, which exists only because λ^3 is readable.

V. THE PATHWAYS, PEAK BY PEAK

We now give every feature of Fig. 1: what makes it possible, how the excitation reaches the emitter, and what would falsify the assignment. Throughout, the delay axis carries a coherence that *never radiates*—it need only be excitable and able to store phase—while the detection axis is constrained by the emitting multipole. That asymmetry does most of the work.

A. Geometry A, cross-polarised

$(q_{\text{AFM}}, E_{12}) = 1.00$: *the magnetoelectric peak; measures W_1^{yy}* . The $H \parallel a$ Zeeman field launches the q_{AFM} magnon, which precesses at 0.90 THz through the delay. It cannot radiate here—it has no c -axis moment—but it need not: it modulates the Tm crystal field through $W_1^{yy} S_y^2 \lambda^1$. Each pulse supplies one magnon leg, so the M_{NL} subtraction keeps the cross term, which rectifies,

$$h_{\text{NL}}^1 = W_1^{yy} A^2 [\cos q_{\text{AFM}} \tau - \cos q_{\text{AFM}}(2t + \tau)] \theta(t), \quad (3)$$

the difference piece being a step switched on at the probe that carries the delay phase but no t dependence. Fed into the E_{12} oscillator it gives $M_{\text{NL}} \propto \cos(q_{\text{AFM}} \tau) [1 - \cos E_{12} t]$: a factorised, rank-one peak. *Why it is possible at all*: an E1-dark magnon is made visible by borrowing the Tm dipole. *Where the energy comes from*: δS_y^2

TABLE I. What each channel detects. An analyser setting selects a polarisation *state*, hence both the magnetic and the electric dipole that radiate into it; there are therefore only *two* detection operators, and the two channels sharing one are constrained to share it exactly. w_{E1} is the $E1/M1$ balance, an $\mathcal{O}(1)$ number our data do not fix (see text).

$(E \parallel a, H \parallel c)$:	$F_z + \mu \lambda^2 + w_{E1}(\lambda^1 + \lambda^3) + \beta \lambda^1 \lambda^2$
A cross $F_z \equiv 0$:	Tm only. $\lambda^{1,2}$ resonant at E_{12} , λ^3 rectified
B same F_z alive but $50\times$ smaller than the Tm terms	
$(E \parallel c, H \parallel a)$:	$F_x + \mu_x(\lambda^5, \lambda^7) + w'_{E1}(\lambda^4, \lambda^6) + d_z^{\text{eff}} \lambda^2$ (no $\lambda^{3,8}$: mirror-odd)
A same F_x ; CEF $\lambda^{4,6}$; $d_z^{\text{eff}} \lambda^2$	
B cross F_x ; Tm part $\equiv 0$ by closure	

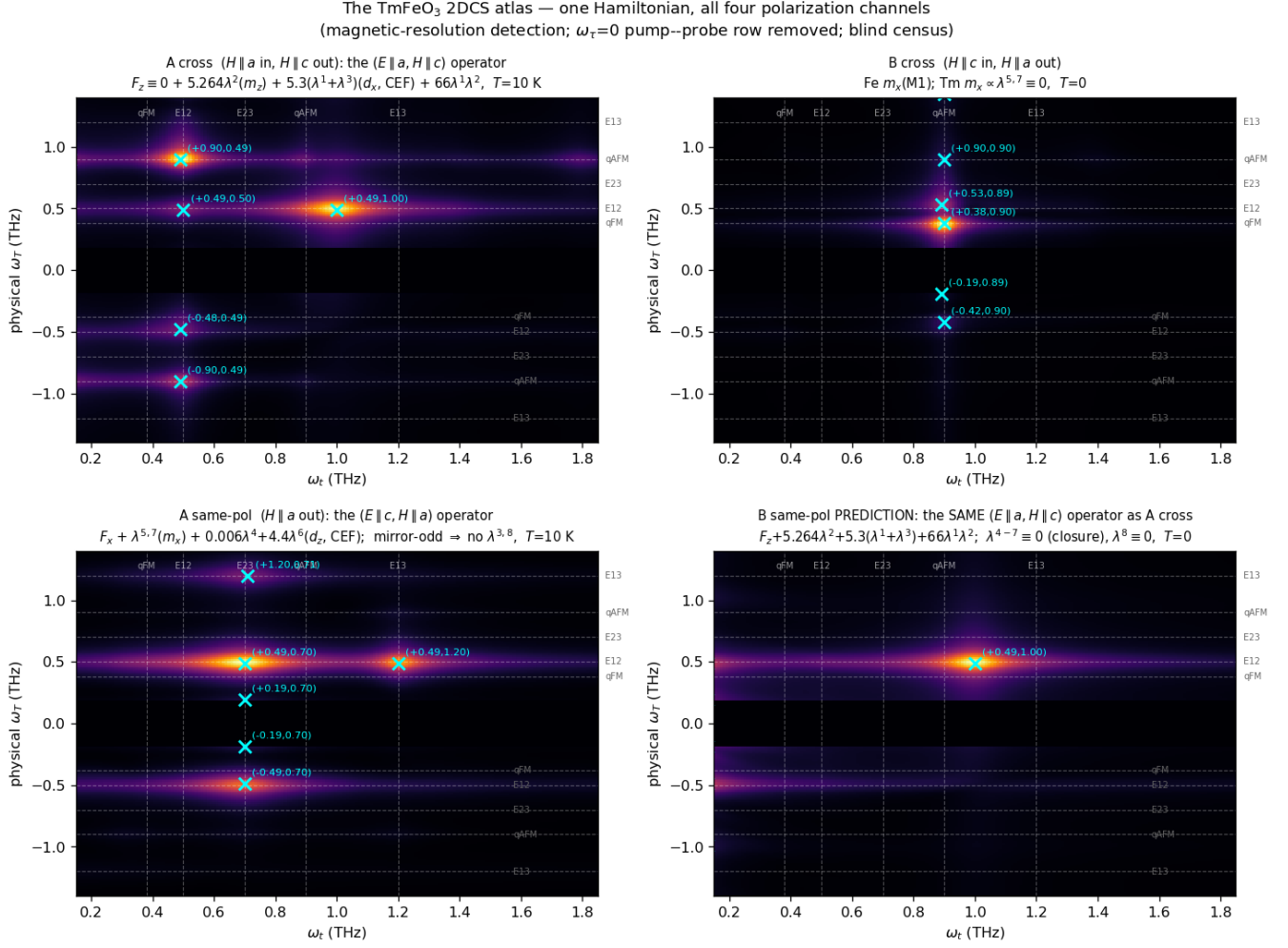


FIG. 1. The atlas: four polarisation channels from one Hamiltonian at one operating point. Dashed guides mark the five mode energies; crosses are blind-census maxima; the $\omega_\tau = 0$ row is removed.

contains only DC and $2q_{\text{AFM}}$, so between free modes this vertex could never emit at ω_{12} ; the quantum is drawn from the pulse edge, an intra-pulse difference-frequency process. This is verified rather than asserted—stretching the pulse from $\sigma = 0.25$ to 1.0 at fixed amplitude collapses the peak by 4.5 decades, while switching off the direct Tm drive changes it by $< 10^{-4}$. *Signature*: $\propto A_{\text{Fe}}^2$, temperature-flat below $\sim 10 \text{ K}$, CEF linewidth along ω_t .

$(E_{12}, 2E_{12}) = 0.97$: *second-harmonic emission; measures β* . The pump stores E_{12} , setting the delay axis at 0.50 ; the detection axis is its second harmonic at $2E_{12} = 1.0000 \text{ THz}$. A moment linear in the qutrit coordinates cannot radiate a harmonic, which is why the model missed this peak until Eq. (2) was included; the $4c$ site's lack of inversion permits it, and Fe at $4b$ is excluded from the mechanism. Evaluated on the trajectories the quadratic channel is remarkably clean, contributing essentially one peak, $5.8\times$ its own (E_{12}, E_{12}) diagonal, identical in local and global sublattice frames. *Discriminators*: the line sits at exactly $2E_{12}$ and tracks

E_{12} , so it does not soften at the spin reorientation as any magnon-derived feature would; it is one field order above the main peak; its width is twice the E_{12} linewidth; and it is independent of W_1^{xz} and of the Zeeman drive.

$(q_{\text{AFM}}, q_{\text{AFM}}) = 0.36$: *the magnon diagonal, seen through the Tm ion*. Since $F_z \equiv 0$ the magnon cannot radiate here at all. What radiates is the Tm dipole, which carries a component at the magnon frequency: in Γ_2 the static vertex $W_1^{xz} S_x S_z \lambda^1$ contains $G_z \delta S_x \lambda^1$, a term *linear* in the magnon fluctuation, which shakes the E_{12} oscillator at 0.90 THz . A driven oscillator responds at the driving frequency, not its own. This is verified in detail: the sideband sits at 0.8999 THz —the magnon, not the $q_{\text{FM}} + E_{12} = 0.876$ combination; its amplitude is *linear* in the Zeeman drive whereas a harmonic would be quadratic; the transfer coefficient $|\lambda^2|_{0.90}/|m_x|_{0.90} = 0.142\text{--}0.155$ is drive-independent over a $20\times$ range, the hallmark of a linear susceptibility evaluated off resonance; and the admixture grows linearly with W_1^{xz} .

$(q_{\text{AFM}}, 2q_{\text{AFM}}) = 0.39$: *prediction*. The same quadratic term must double *any* frequency present in the Tm dipole, so the magnon-forced component radiates its harmonic at 1.79 THz. Observing it confirms Eq. (2) independently of the $2E_{12}$ peak; its absence with $(E_{12}, 2E_{12})$ present would falsify that equation.

$(-q_{\text{AFM}}, E_{12}) = 0.52$ and $(\pm E_{12}, E_{12}) = 0.33$. The first is not a separate mechanism: the W vertex is symmetric under the two time orderings of the magnon coherence, so the signal appears on both signed sides with the non-rephasing branch stronger. The second is the one-quantum diagonal, the directly driven E_{12} read through the same dipole; its size relative to the cross peaks is a sensitive fluence diagnostic, falling from 1.45 at $A_{\text{Fe}} = 1.2$ to 0.16 at 0.12, which is how the operating point was fixed.

B. Geometry B, cross-polarised

Here the analysed moment is m_x , which the q_{AFM} magnon carries directly, so both peaks emit at 0.90 THz through the Fe magnetic dipole and the Tm ion is radiatively absent.

$(q_{\text{FM}}, q_{\text{AFM}}) = 1.00$: *magnon-magnon; measures the Fe sector alone*. $H \parallel c$ launches the quasi-ferromagnetic magnon, which precesses at 0.38 through the delay; single-ion anisotropy and the Dzyaloshinskii-Moriya interaction couple the branches, so the stored q_{FM} phase is re-emitted on q_{AFM} . No Fe-Tm coupling enters at any vertex, which is why this peak is temperature-flat: a pure two-magnon process, independent of the qutrit populations.

$(E_{12}, q_{\text{AFM}}) = 0.45$: *the gated transfer; measures κ_{1y}^B* . $E \parallel a$ stores a Tm E_{12} coherence—parity-allowed here, since $1 \leftrightarrow 2$ is owned by this polarisation—and a Fe-Tm vertex transfers it to the magnon. Isolating the vertices settles which: with κ_{1y}^B alone the peak is present, with the static hybridisation alone it is absent, and *no* strength of the static vertex recovers it. A sweep over $W_1^{xz} \in \{0.01, \dots, 0.09\}$ shows the static vertex never creates a second resolved peak; instead the two delay rows coalesce and the leading peak migrates ($0.38 \rightarrow 0.40 \rightarrow 0.48$ THz) while hybridisation-shifted E_{12} emission leaks in at 0.44–0.45 THz. This is the structural signature of a mixer: a static vertex *moves* peaks (level repulsion, which also shifts modes off their 1D-measured positions), whereas a gated converter *adds* a peak at unshifted lines. The data show two resolved peaks at the unshifted energies, so the gate is required. Physically κ^B supplies the $0.50 \rightarrow 0.90$ THz mismatch impulsively from the photon while $B_z(t) \neq 0$ —and the same gate makes it *identically silent* in geometry A, where its background would otherwise contaminate the clean W_1^{yy} peak. *Signature*: falls as the population difference $n_1 - n_2$ ($0.46 \rightarrow 0.45 \rightarrow 0.40 \rightarrow 0.36$ at 0/5/8/10 K), the sharpest thermal discriminator in the atlas against the temperature-flat magnon-magnon peak.

$(-q_{\text{FM}}, q_{\text{AFM}}) = 0.17$, $(q_{\text{AFM}}, q_{\text{AFM}}) = 0.14$,

$(E_{12} + q_{\text{AFM}}, q_{\text{AFM}}) = 0.07$. The rephasing partner; the q_{AFM} diagonal, which here is genuine Fe M1 emission unlike its geometry-A counterpart; and a combination tone, the sum-frequency of the two stored coherences radiating on the magnon line, which is W -fed and so offers an independent handle on the W scale.

The post-drive buildup. The measured geometry-B signal continues to grow after the pulse. No coherent vertex can do this once the measured linewidths are enforced: because $B \parallel c$ never drives q_{AFM} directly, all geometry-B magnon amplitude is *converted*, and a converted signal inherits its source’s lifetime—with $T_2(E_{12}) = 8$ ps everything is over within ~ 10 ps. Even the τ -labelled population grating fails, because its delay label is written by the E_{12} coherence and is therefore T_2 -confined. The buildup requires the one reservoir outliving every T_2 : the CEF populations, fed incoherently by the dissipated pulse energy and acting on the magnon through $W_3 S_\alpha S_\beta \lambda^3$ —the mechanism of thermally driven spin reorientation. Implemented in the solver, it reproduces the observation: the $\omega_\tau = E_{12}$ row falls off the impulsive peak, recovers on the repopulation time, and persists beyond 100 ps, 50–100× above the coherent-only model.

C. Geometry A, same-polarised

The analysed moment is m_x , and three contributions reach it, each required by symmetry and each carrying measured features: the Tm CEF moments (emitting at E_{13} , E_{23}), the Fe F_x (magnon frequencies), and $d_z^{\text{eff}} \lambda^2$ (emitting at E_{12}), the last demanded by reciprocity once d_z^{eff} appears in the drive. Fitting those three weights to the digitised measurement, with the cross channels held fixed, gives Table II.

The z -dark $1 \rightarrow 3$ dipole—measured, not assumed. The measured map contains delay frequencies $\omega_\tau \in \{E_{12}, q_{\text{AFM}}\}$ only. A bright $1 \rightarrow 3$ dipole would store an E_{13} coherence at first order and produce an (E_{13}, E_{23}) peak with *exactly* the same dipole product $\mu_{12}\mu_{13}\mu_{23}$ as the observed (E_{12}, E_{23}) , so no parameter choice could separate them: the absence of the whole row is a selec-

TABLE II. Geometry-A same-polarised channel: measured amplitudes against the model with the three symmetry-required detection weights fitted. The μ_{13}^2 weight fits to zero.

peak	exp.	model	pathway
$(E_{12}, 0)$	1.00	—	rectified storage
(E_{12}, E_{23})	1.00	1.00	coherence→pop.→ ρ_{23}
(E_{12}, q_{FM})	0.45	0.38	Tm→magnon, q_{FM}
(E_{12}, q_{AFM})	0.40	0.46	Tm→magnon, q_{AFM}
$(0, E_{12})$	0.40	—	pump-probe row
(q_{AFM}, E_{12})	0.30	0.32	magnon→Tm via d_z^{eff}
(q_{AFM}, E_{13})	0.30	0.09	magnon delay, CEF emission
(E_{12}, E_{13})	0.15	0.44	single-action transfer

tion rule. It does not contradict the measured E_{13} oscillator strength ($f = 8.7$), which lives in the orthogonal polarisations—the same transition is bright in one polarisation and dark in another, a fact the 2D map resolves and 1D absorption, summed over its bright polarisations, does not. In the fit the residual admixture is not imposed; it comes out at zero.

$(E_{12}, E_{23}) = 1.00$: *the dominant transfer, via a population.* With the single-action route closed by the dark dipole, the surviving pathway acts twice: one probe interaction converts the stored E_{12} coherence into a level-2 population that still carries the $\omega_\tau = E_{12}$ phase, and a second converts that population into the emitting E_{23} coherence. Being co-rotating it lands non-rephasing, the observed side, and it involves μ_{13} at no vertex—which is why it survives while its E_{13} partner is throttled.

$(E_{12}, 0) = 1.00$: *the rectified line.* Taken at its zero-frequency output the same double action rectifies the stored coherence into a slow net magnetisation of the detection window. This is the strongest measured feature; in the 2D maps it is suppressed by construction, since the $t \geq 3$ window with Gaussian apodisation removes $\omega_t \rightarrow 0$, so it is analysed separately through $S_{\text{DC}}(\tau) = \langle M_{\text{NL}} \rangle_t$, where it appears as a sharp line at $+0.49$ THz. Its absence from Table II is an analysis-window statement, not a physical one.

$(E_{12}, q_{\text{FM}}) = 0.45$ and $(E_{12}, q_{\text{AFM}}) = 0.40$: *Tm $\rightarrow$ magnon, read through Fe.* The stored E_{12} coherence is handed to the Fe sublattice by the static couplings and radiates through F_x , which carries both branches. These are the reverse of the geometry-A cross peak—there the magnon feeds the Tm dipole, here the Tm coherence feeds the magnon—so the static W family is read in both directions. They require the Fe term in the detection operator; without it this channel has no magnon-frequency emission at all.

$(q_{\text{AFM}}, E_{12}) = 0.30$: *the reciprocity partner, observed.* The magnon occupies the delay axis and the Tm ion radiates at E_{12} into the *same*-polarised channel. That emission is possible only through d_z^{eff} —and reciprocity demands it, because the same d_z^{eff} is what lets $E \parallel c$ create the E_{12} coherence in the first place. We had earlier reported this partner as absent and treated the mismatch as an open tension; the digitised map shows it present at 0.30 of maximum, and the fitted weight reproduces it at 0.32. This peak is the direct measurement of d_z^{eff} .

$(q_{\text{AFM}}, E_{13}) = 0.30$ and $(E_{12}, E_{13}) = 0.15$. The first has the magnon on the delay axis, handed to the Tm ion and emitted at E_{13} ; the second is a single probe action rotating the bra of the stored ρ_{12} into ρ_{13} , again co-rotating, with its emission throttled by the nearly dark μ_{13}^z . These are the two the model gets wrong, and Sec. VI shows they fail together.

Two model features not in the measured list, both predictions. The *rephasing remnant* ($-E_{12}, E_{23}$) at 0.67: checked against a standalone exact-qudit integrator, where the rephasing partner retains 0.47–0.58 of the dominant peak across all drive strengths, dipole phases and

dampings—an exact floor for any three-level ion under this pulse. If the measured map truly contains nothing there, the suppression must come from outside the point-ion model: quadrant folding in the display convention, or propagation in a thick sample. And the *two-quantum row* (E_{13}, E_{23}) at 0.32: an E_{13} delay coherence does survive the dark dipole, but only at second order, the pump driving $1 \rightarrow 2$ and $2 \rightarrow 3$ in succession rather than $1 \rightarrow 3$ directly. Its weakness relative to the first-order row a bright μ_{13}^z would give is exactly why the model is consistent with an inventory listing no E_{13} delay features.

D. Geometry B, same-polarised: a prediction and a β meter

This channel is analysed with the *same* operator as geometry-A cross, so it is a prediction with no freedom left. Calling it an “ m_z readout” would understate it: $\mu\lambda^2$ is the Tm E_{12} CEF moment and is $\sim 50 \times$ the Fe F_z term, so the channel is CEF-dominated, not magnon-dominated, and it additionally carries the electric λ^1 and—uniquely among the four channels—the population readout λ^3 . What vanishes here is only λ^{4-7} , by the sub-algebra closure, so no E_{13} or E_{23} emission is possible; λ^8 vanishes for the same reason.

Two features follow. The map is led by the second harmonic $(E_{12}, 2E_{12})$, because λ^2 is driven *directly* through μ_{12}^z and the $\beta \simeq 7\mu$ inferred in geometry A then makes the harmonic $4.1 \times$ the magnon–magnon feature; measuring it reads β off directly. Beneath it, at 0.53 of that peak, sits a *rectified* pair at $(\pm E_{12}, \omega_t \rightarrow 0)$: the drive writes a level population modulated at the delay frequency E_{12} , and λ^3 is readable in this polarisation and no other. This is a direct optical readout of the population channel W_3 —the same quantity whose slow transfer produces the post-pulse buildup of geometry-B cross—and it is the only place in the atlas where the $E1/M1$ balance matters, since with the magnetic moment alone it would fall to 0.09. A measured rectified line at $(E_{12}, 0)$ in this channel therefore measures w_{E1} ; its absence would mean d_x has no permanent component despite the mirror allowing one.

VI. THE ONE OPEN ITEM

The two failures in Table II are not independent. Both peaks are detected at E_{13} and therefore share an emission weight, so their ratio is a property of the conversion dynamics alone:

$$\frac{(E_{12}, E_{13})}{(q_{\text{AFM}}, E_{13})} : \quad \text{model 4.9,} \quad \text{experiment 0.5.} \quad (4)$$

No choice of detection weights can move it, and no rescaling of the drive does either, since both routes are first order in the coherence they start from. The

model routes too much E_{13} emission through the E_{12} -excitation row relative to the magnon row—equivalently, the magnon→Tm handoff feeding (q_{AFM}, E_{13}) is too weak relative to the intra-Tm $\rho_{12} \rightarrow \rho_{13}$ rotation. The natural places to look are the strength of that handoff at the E_{13} block, which our vertex set feeds only through λ^1 , and a λ^4 -targeting vertex, allowed in the W family but so far held at zero.

VII. FALSIFIABLE CONTENT

Fifteen independent observations constrain five couplings (W_1^{yy} , W_1^{xz} , κ_{1y}^B , W_3 , β), one selection rule and two drive parameters. The sharpest tests: any peak emitting at q_{AFM} inherits the $15\times$ sharper magnon linewidth; $(q_{\text{FM}}, q_{\text{AFM}})$ is temperature-flat while (E_{12}, q_{AFM}) falls as $n_1 - n_2$; rotating either polarisation away from c must revive the $\omega_\tau = E_{13}$ rows; the $2E_{12}$ line tracks E_{12} and so does *not* soften at the spin reorientation, whereas every feature relying on $d_z^{\text{eff}} \propto \langle F_x \rangle$ —the E_{12} -excitation family and the (q_{AFM}, E_{12}) emission peak—must vanish there; the geometry-B same-polarised channel should show the harmonic peak led, with a rectified $(E_{12}, 0)$ line at ~ 0.5 of it whose presence measures the electric population readout w_{E1} ; and adiabatically long pulses collapse the geometry-A cross peak. The temperature scan through the reorientation is the decisive one, since it separates the symmetry-broken from the conversion-generated features

and measures d_z^{eff} by the size of the collapse.

VIII. CONCLUSION

Two symmetry-selected Fe–Tm vertices, one Fe-sector magnon–magnon coupling, a mirror-parity rule assigning each polarisation to one qutrit block, and a hyperpolarisability allowed by the absence of inversion at the rare-earth site account for the measured 2DCS atlas of TmFeO_3 , computed from measured inputs with no free lineshape parameters and one residual dynamical ratio. Every peak has been given a pathway, every structural absence a selection rule, and every assignment a falsifiable signature. The program—bound the vertex space by symmetry, assign every feature and every absence, and let the atlas overdetermine the model—applies to any magnet whose 2D map spans distinct subsystems, and turns 2DCS from a spectroscopy of modes into a spectroscopy of Hamiltonian terms.

ACKNOWLEDGMENTS

Derivations, numerical protocol and the complete falsification ledger are in the Supplemental Material; the long-form note and all run recipes accompany the data.

-
- [1] J. Lu, X. Li, H. Y. Hwang, B. K. Ofori-Okai, T. Kurihara, T. Suemoto, and K. A. Nelson, *Phys. Rev. Lett.* **118**, 207204 (2017).
 - [2] S. Mukamel, *Principles of Nonlinear Optical Spectroscopy* (Oxford University Press, New York, 1995).
 - [3] T. Yamaguchi, *J. Phys. Chem. Solids* **35**, 479 (1974).
 - [4] A. V. Kimel, A. Kirilyuk, A. Tsvetkov, R. V. Pisarev, and Th. Rasing, *Nature* **429**, 850 (2004).
 - [5] *Resolving the spin reorientation and crystal-field transitions in TmFeO_3 with terahertz transient*, *Sci. Rep.* **6**, 23648 (2016).
 - [6] *Terahertz magnon-polaritons in TmFeO_3* , *ACS Photonics* **5**, 1375 (2018).